

Executive MBA

Financial Accounting

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Lecture 4: Financial Reporting for Long Term Liabilities



The University of Chicago Booth School of Business

Topics

- Present Value and Future Value Concepts.
- Financial Reporting for Long term Liabilities:
 - We will focus on bond valuation and reporting.

Long-Term Asset and Liability Valuation

- Because of the *time value of money*, interest is a common feature of delayed payment liabilities and represents the price charged for delaying payment:
 - To capture the cost of borrowing, long-term *monetary liabilities* should be measured on the financial statements at the *present value of the future cash flows* required to satisfy the obligation.
 - In general, both long-term monetary assets (e.g., notes receivable) and liabilities (notes payable) are presented on the balance sheet at the *present value* of future cash inflows or outflows, respectively.
 - However, current monetary assets and liabilities (short-term) are not discounted because of the short-term nature of their maturity (i.e., the amount of discount is immaterial).

Time Value of Money: Future Value

- The *future value* of money is concerned with the following question: How much is money paid or received today worth *in the future*?

Example: Suppose you invest \$100 in a project that pays a 10% annual interest (i.e., interest is compounded annually). What would the balance in the account be at the end of 3 years?

- Stated differently, what would the *future value (FV)* of \$100 be 3 years from today if it earns a rate of return of 10% compounded annually?
 - $FV = \$100 \times [1+0.1] \times [1+0.1] \times [1+0.1] = 100 \times (1.1)^3 = \133.10
 - In general, $FV = PV \times (1 + r)^n$ where
 - PV = present value of amount today
 - r = interest rate per period
 - n = number of periods

Time Value of Money: Present Value

- The *present value* of money is concerned with the following question: How much is money paid or received in the *future* worth *today*?

Example: How much do you have to invest today in an account that pays 10% interest compounded annually in order to have \$100 at the end of 3 years?

- Stated differently, what is the present value (PV) of \$100 received 3 years from today from a project that earns a rate of return of 10% compounded annually?

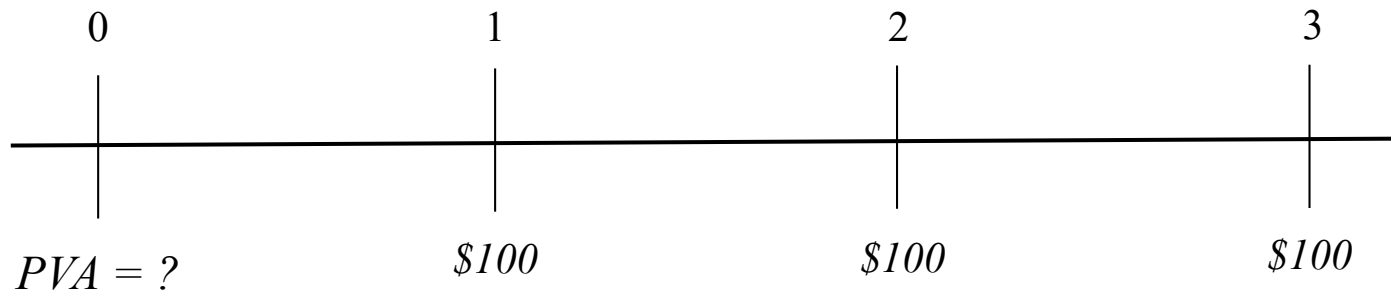
➤ $PV \times (1.1)^3 = \$100 \Rightarrow PV = \$100 / (1.1)^3 = \$75.13$

➤ In general, $PV = FV / (1 + r)^n$ where:

- PV = present value of amount today.
- FV = Amount to be received or paid in the future.
- r = interest rate per period.
- n = number of periods.

Time Value of Money: Present Value of an Annuity

Example: Suppose you want to receive \$100 at the end of *each* of the next 3 periods, how much do you need to invest today (say at date 0) in an investment that pays a rate of return of 10% compounded annually? That amount is called the present value of an *ordinary (PVA) annuity* of \$100 discounted for 3 periods at a rate of return of 10% per period.



- $$\begin{aligned} PVA &= 100/(1+0.10) + 100/(1+0.10)^2 + 100/(1+0.10)^3 \\ &= 100 \times [1 - (1 + 0.1)^{-3}]/10\% = \$248.69 \\ &= 100 \times 2.4869, \text{ where } 2.4869 \text{ is the } PV \text{ of an ordinary} \\ &\quad \text{annuity factor (PVANN) for 3 periods discounted at} \\ &\quad 10\% \text{ per period} \end{aligned}$$

Present Value of an Ordinary Annuity

- In general,

$$PVANN = \frac{[1 - (1 + r)^{-n}]}{r}$$

- Can we derive the above factor from first principles?
 - Consider an annuity stream that pay \$1 at the end of each of the next n periods compounded at an annual rate of return $r\%$ per period. What is the present value of that stream of payments?

Bond Terminology

- A **bond** is a financial instrument that represents a formal promise to repay both the amount borrowed as well as any interest rate on the amount borrowed. The precise terms of the borrowing are specified in a document called a **bond indenture agreement**.
- **Coupon bond** is a bond that pays interest periodically and also repays the principal amount at maturity.
- **Serial bond** is a bond that requires not only periodic payments of interest *but also* a portion of the principal *throughout* the life of the bond (similar to lease or mortgage payments).
- **Zero-coupon bonds** provide no periodic payments of interest while the bond is outstanding but then repays *all* principal and interest amounts at maturity.
- **Convertible bonds** permit the holder to exchange the bonds for shares of the firm's common stock under certain conditions.

Bond Terminology, cont'd

- **Secured vs. unsecured bonds:** Secured bonds are those that are collateralized by an asset, such as property, equipment or another income stream, while unsecured bonds are not secured by any specific asset, but rather by the "full faith and credit" of the issuer. Unsecured bonds are also called **debenture bonds**.
- **Consol bond** is a government bond with no maturity. A consol bondholder, in exchange for an initial payment, receives a steady stream of coupon payments until the government opts to redeem or “call” the bond.
- **Face value, principal value, or par value:** The amount due at maturity.
 - Face value = maturity value for coupon and zero-coupon bonds but not for serial bonds.

Bond Terminology, cont'd

- **Interest rate:**

- **Coupon rate or nominal rate or stated rate:** The contractual interest rate specified in the *bond indenture agreement* that determines the periodic interest payment amount. For a zero-coupon bond, the coupon rate is 0% by definition.
 - Interest payments by the borrower is computed by multiplying the *face value* of the bond by the stated *coupon rate*.
- **Effective yield or initial yield:** The interest rate that market participants use to price the bond on its *date of issuance*:
 - Reflects the *effective interest rate* from the perspective of the borrower and therefore the effective rate of return earned by bondholders.
- **Market interest rate:** The prevailing marketplace rate at *any* date subsequent to the date of the initial offering for a comparable piece of debt (maturity and risk).
 - Captures the rate of return demanded by market participants on an instrument at *any* point in time.

Interest Rate, cont'd

Consider a \$10M 4% 30-year bond issued on June 30, 2024, due on June 30, 2054. At issuance, the bonds are priced to yield 6% annually. On June 30, 2025, the market interest rate on bonds of the same risk and maturity is 5%.

What is the coupon rate, initial yield (effective interest rate), and market rate on the bond at issuance on June 30, 2024, and one year after issuance (i.e., on June 30, 2025)?

	June 30, 2024	June 30, 2025
Coupon rate		
Initial yield		
Market rate		

Two Methods to Value Bonds

1. **Amortized cost:** Use the initial yield (effective interest rate) to discount all remaining future cash outflows (at issuance *and* at any point in time after issuance).
 - The carrying value of the bond at *any* point in time is therefore the *PV* of all future cash outflows discounted at the *initial yield*. In practice, the *net proceeds* from issuing the bond determines the initial yield.
2. **Fair value:** Use the *current market interest rate* to discount all future cash outflows.
 - The carrying value of the bond is therefore always *equal* to the market price or fair value of the bond at any point in time.
- Regardless of the valuation method, the carrying value of the bond *at issuance* is the same under both methods and is equal to the market or issuance price of the bond. However, *after issuance*, the carrying values would be different for method (1) vs. (2).
- Companies have an option to use either method (1) or (2) to value long-term debt on their financial statements. Most bond issuers (borrowers) choose (1) for recognition on the balance sheet and may disclose fair values in their financials.

Coupon rate vs. Initial yield: Why are they different?

1. Credit market conditions and/or borrower creditworthiness may have changed resulting in a difference between market interest rate at issuance and coupon rate:
 - a. If coupon rate = market interest rate at issuance, bonds are priced at face value so that initial yield = coupon rate.
 - b. If coupon rate < market rate at issuance, bonds are priced at a discount so that initial yield > coupon rate.
 - c. If coupon rate > market rate at issuance, bonds are priced at a premium so that initial yield < coupon rate.
2. Debt Issuance costs (transaction cost) reduce the proceeds from issuance of debt so that initial yield > coupon rate. Bond is priced below par and therefore issued at a discount.
 - Note therefore that *even if* coupon rate = market interest rate at issuance, initial yield > coupon rate due to debt issue costs so that debt is issued at a discount.
 - If bonds are already issued at a discount due to 1(b), then debt issue costs *further* increase the discount. If issued at a premium due to 1(c), then debt issue costs reduce the premium and if the issue costs are large, the bonds may be issued at a net discount.

Alibaba Unsecured Notes

The following table provides a summary of the Company's unsecured senior notes as of March 31, 2022 and 2023:

Principal or Face value

	As of March 31,		Effective interest rate
	2022	2023	
	RMB	RMB	
	(in millions)		
<i>Coupon rate</i> <i>Maturity date</i>			
US\$700 million 2.800% notes due 2023	4,439	4,800	2.90%
US\$2,250 million 3.600% notes due 2024	14,256	15,410	3.68%
US\$2,550 million 3.400% notes due 2027	16,091	17,397	3.52%
US\$1,500 million 2.125% notes due 2031	9,469	10,234	2.20%
US\$700 million 4.500% notes due 2034	4,400	4,754	4.60%
US\$1,000 million 4.000% notes due 2037	6,300	6,807	4.06%
US\$1,000 million 2.700% notes due 2041	6,256	6,761	2.80%
US\$1,750 million 4.200% notes due 2047	11,014	11,898	4.25%
US\$1,500 million 3.150% notes due 2051	9,448	10,206	3.19%
US\$1,000 million 4.400% notes due 2057	6,290	6,795	4.44%
US\$1,000 million 3.250% notes due 2061	6,296	6,803	3.28%
Carrying value	94,259	101,865	
Unamortized discount and debt issuance costs	668	665	
Total principal amounts of unsecured senior notes	94,927	102,530	
Less: current portion of principal amounts of unsecured senior notes	—	(4,801)	
Non-current portion of principal amounts of unsecured senior notes	94,927	97,729	

The effective interest rates for the unsecured senior notes include the interest charged on the notes as well as amortization of the debt discounts and debt issuance costs.

Alibaba Unsecured Notes

As of March 31, 2023, the future principal payments for the Company's unsecured senior notes will be due according to the following schedule:

	Principal amounts
	RMB
	(in millions)
Within 1 year	4,801
Between 1 to 2 years	15,431
Between 2 to 3 years	–
Between 3 to 4 years	–
Between 4 to 5 years	17,488
Thereafter	64,810
	102,530

As of March 31, 2022 and 2023, the fair values of the Company's unsecured senior notes, based on Level 2 inputs, were US\$14,067 million (RMB89,319 million) and US\$12,523 million (RMB85,886 million), respectively.

Bond Accounting: An Illustration

- Firm sells a \$1,000 6% bond due 30/6/2030 to market investors on 30/6/2024. Interest on the bond is *compounded and paid semi-annually* in cash. The market prices the bond at \$940 at issuance and firm pays debt issue costs of \$20.

Requirements:

1. What are the net proceeds (net of debt issue costs) from issuing the bonds at 30/6/2024?
2. What is the initial annual yield or effective annual interest rate on the bond?
3. Show the journal entries to recognize (1) issuance of the bond, (2) interest expense recognized on 31/12/2024 and on 30/6/2025.

Determining the initial yield (effective interest rate)

- Market Price of bond = \$940 so that the original issue discount = \$60.
- Bond issuance costs = \$20 so that the net bond cash proceeds at issuance = Face value – original issue discount – bond issuance costs = \$1,000 – 60 – 20 = \$920.
- Let initial semi-annual yield = r , then r satisfies the following equation:

$$920 = \sum_{n=1}^{12} \frac{30}{(1+r)^n} + \frac{1,000}{(1+r)^{12}}$$
- Using the IRR function in Excel, $r = 3.84\%$ or annual yield 7.68% (See Excel file on Canvas):

	30/6/2024	31/12/2024	30/6/2025	31/12/2025	30/6/2026	31/12/2026	30/6/2027	31/12/2027	30/6/2028	31/12/2028	30/6/2029	31/12/2029	30/6/2030
Payments	-920	30	30	30	30	30	30	30	30	30	30	30	1030
IRR(B5:N5)	3.84%												

Bond Example continued...

Recognizing issuance of bond at 30/6/2024:

Cash	\$920	
Bond Discount (contra-liability)	80	
Bond Payable		\$1,000

Recognizing interest expense for 6 months ended 31/12/2024:

Interest Expense	\$35.4 (initial yield x $CV_{30/6/2024} = 3.84\% \times \\920)	
(2) Cash		\$30.0 (= $3\% \times 1,000$)
(2) Bond Discount (derived)		5.4

Carrying value of bond at 31/12/2024: $\$920 + 5.4 = \925.4

Recognizing interest expense for 6 months ended 30/6/2025:

Interest Expense	\$35.6 (initial yield x $CV_{31/12/2024} = 3.84\% \times \\925.4)	
(2) Cash		\$30.0 (= $3\% \times 1,000$)
(2) Bond Discount (derived)		5.6

Bond Amortization Table

Period	Beginning Carrying Value	Interest Expense = Beginning CV X Initial yield of 3.84%	Cash Interest = Face Value X Coupon rate of 3%	Discount Amortized = Interest Expense – Cash Interest	Ending Carrying Value = Beginning Carrying Value + Discount Amortized
1	\$920.0	\$35.4	\$30.0	\$5.4	\$925.4
2	925.4	35.6	30.0	5.6	931.0
3	931.0	35.8	30.0	5.8	936.7
4	936.7	36.0	30.0	6.0	942.8
5	942.8	36.2	30.0	6.2	949.0
6	949.0	36.5	30.0	6.5	955.5
7	955.5	36.7	30.0	6.7	962.2
8	962.2	37.0	30.0	7.0	969.2
9	969.2	37.3	30.0	7.3	976.5
10	976.5	37.5	30.0	7.5	984.0
11	984.0	37.8	30.0	7.8	991.9
12	\$991.9	\$38.1	\$30.0	\$8.1	\$1,000.0
Cumulative		\$440.0	\$360.0	\$80.0	

Consider the following bonds issued on 31/12/2024...

1. \$1,000 0% bond due 31/12/2026 priced to yield 10% and interest compounded annually.
2. \$1,000 6% bond due 31/12/2026 priced to yield 6% and interest compounded and paid annually.
3. \$1,000 6% bond due 31/12/2026 priced to yield 10% and interest compounded and paid annually.
4. \$1,000 6% bond due 31/12/2026 priced to yield 4% and interest compounded and paid annually.

Show all the journal entries associated with each of the above bonds issued by a borrower from the date of issuance until maturity under the *Amortized Cost method*.

Example 1

\$1000 0% bond due 31/12/2026 priced to yield 10% and interest compounded annually.

- Market price of bond at issuance = $\$1,000/(1.1)^2 = \826
- Entry at 31/12/2024 to recognize proceeds from issuance:

(1) Cash	\$826	
(1) Bond Discount	174	
(1) Bond Payable		\$1,000

- Carrying value of bond at issuance = Face value – bond discount
= $\$1,000 - 174 = \826
- The bond discount of \$174 represents the *total cost of borrowing* (or *cumulative interest expense*) and is allocated to interest expense during fiscal 2025 and 2026.

Example 1 cont'd

- Interest expense for fiscal 2025: $\$826 \times 0.10 = \82.6 or \$83

(2) Interest Expense	\$83	
(2) Bond Discount		\$83

- Carrying value of the bond at 31/12/2025 = $\$826 + 83 = \909 .
- Unamortized discount at 31/12/2025: $\$174 - 83 = \91
- Interest expense for fiscal 2026: $\$909 \times 0.10 = \90.9 or \$91

(3) Interest Expense	\$91	
(3) Bond Discount		\$91

- Unamortized discount at 31/12/2026: $\$91 - 91 = \0
- Carrying value of bond at 31/12/2026 = $\$909 + 91 = \$1,000$ = face value which is paid off at maturity, i.e., at 31/12/2026:

(4) Bond Payable	\$1,000	
(4) Cash		\$1,000

Example 1 Illustrated via T-Accounts

Bond Payable

Bond Discount

	(1) \$1,000 (31/12/24)	(1) \$174 (31/12/24)	
	\$1,000 (31/12/25)		(2) \$83 (31/12/25)
(4) \$1,000 (31/12/26)		\$91 (31/12/25)	
	\$0 (31/12/26)		(3) \$91 (31/12/26)
		\$0 (31/12/26)	

Interest Expense (Retained Earnings)

Cash

(2) \$83 (31/12/25)		(1) \$826 (31/12/24)	
(3) \$91 (31/12/26)			(4) \$1,000 (31/12/26)
			\$174

*Note that cumulative interest expense of \$83 + \$91 = \$174
= net cash outflows of \$174*

Example 1 demonstrates that...

- Interest expense, which captures the cost of borrowing, is still recognized even though the coupon rate on the bond is zero.
- In general, interest expense in a period under the *Amortized Cost Method* is calculated using the *Effective Interest Method*:

$$\text{Interest expense} = \text{CV} \times \text{Effective interest rate}$$

Where effective interest rate = initial yield and CV = carrying value of the bond at the beginning of the period.

- In general, whenever a bond is issued at a discount (initial yield > coupon):

$$\text{Interest expense} = \text{coupon interest} + \text{discount amortized}$$

- However, when a bond is issued at a premium (initial yield < coupon):

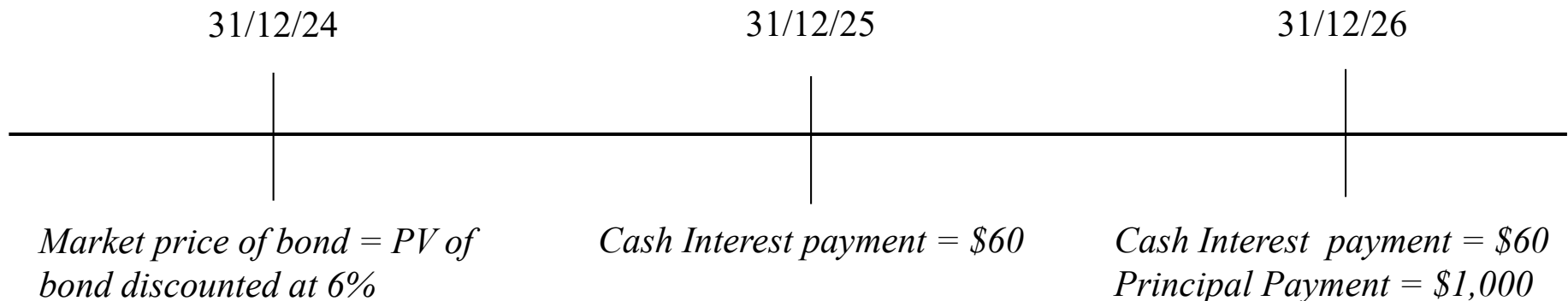
$$\text{Interest expense} = \text{coupon interest} - \text{premium amortized}$$

Example 2

\$1,000 6% bond due 31/12/2026 priced to yield 6% and interest compounded and paid annually.

- Market price of bond at issuance = *PV of all future cash flows* from the bond discounted at the *initial yield* of 6%.
- Interest paid in 2025 and 2026 = coupon rate x face value

$$= 6\% \times \$1,000 = \$60$$



Example 2 cont'd

- Market Price = PV of interest payments discounted at the market yield of 6% + PV of principal discounted at the market yield of 6%:

Present Value Annuity factor for 2 periods, 6% interest rate per period *Present Value factor for 2 periods, 6% interest rate per period*

$$\begin{aligned} \text{Price of bond} &= \$60 \times \overset{\downarrow}{PVANN}(2,6\%) + \$1,000 \times \overset{\downarrow}{PV}(2,6\%) \\ &= \$60 \times 1.83 + 1,000 \times 0.89 = 110.0 + 890.0 = \$1,000 \end{aligned}$$

- Why is the issue price of the bond of \$1,000 in Example 2 (unlike the issue price in Example 1) *equal* to its face value of \$1,000?
- When the **bond coupon rate = initial yield**, then the ***bond issue price = face value***.
 - Therefore, there is *no discount or premium* associated with such a bond, thus—until the bond is paid off at maturity—*its carrying value always equals its face value, which in turn always equals its PV*.
 - Therefore, interest expense = coupon interest every period.

Example 2 cont'd

- Entry at 31/12/2024 to recognize bond proceeds from issuance:

Cash	\$1,000	
Bond Payable		\$1,000

- Interest expense for 2025 *and* 2026 = carrying value x initial yield
= 1,000 x 0.06 = \$60
= face value x coupon rate
- Entries made at 31/12/2025 and 31/12/2026 to recognize interest payments:

Interest Expense	\$60	
Cash		\$60

- Finally, at 31/12/2026, the face value of the bond is paid off:

Bond Payable	\$1,000	
Cash		\$1,000

Example 3

\$1,000 6% bond due 31/12/2026 with an initial annual yield of 10% and interest compounded and paid annually.

- Coupon interest = \$ 1,000 x 6% = \$60 every year
- Issue price of bond = $\$60 \times PVANN(2,10\%) + \$1,000 \times PV(2,10\%)$
= $\$60 \times 1.74 + 1,000 \times 0.8265$
= $104.0 + 826.5 = \$930.5$
- Bond is issued at a discount of \$69.5, Why?
- When *initial yield* > *coupon rate*, the bond's issue price is *less* than its face value!
- Carrying or book value of bond = face value – unamortized discount

(1) Cash	\$930.5	
(1) Bond Discount	69.5	
(1) Bond Payable		\$1,000

Example 3 cont'd

- *Cumulative interest expense* recognized by the firm over 2 years = bond discount of \$69.5 + total coupon interest payments of \$120 = \$189.5.
- Interest expense = coupon interest + discount amortized, so that when a bond is issued at a discount, ***interest expense recognized exceeds the coupon interest payment by the amount of discount amortized!***

(2) Interest Expense [2025]	\$93.1 (= effective yield x CV = 10% x \$930.5)	
(2) Cash		\$60.0 (= 6%*1,000)
(2) Bond Discount (plug)		33.1

- Unamortized discount at 31/12/2025 = \$69.5 – 33.1 = \$36.4 and carrying value of bond at 31/12/2025 = \$930.5 + 33.1 = \$963.6

(3) Interest Expense [2026]	\$96.4 (= 10%*963.6)	
(2) Cash		\$60.0 (= 6%*1,000)
(2) Bond Discount (plug)		36.4

Example 3 cont'd

- Note the unamortized discount at 31/12/2026, just after the second and final interest payment, $\$36.4 - 36.4 = \0 .
- Carrying value of bond at 31/12/2026 = $\$963.6 + 36.4 = \$1,000$ = face value, which is paid off at maturity 31/12/2026 as follows:

(4) Bond Payable	\$1,000	
(4) Cash		\$1,000

- Interest expense can also be calculated using the following *bond discount amortization table*:

	(a)	(b)	(c)	(d)	(e)
Year	Beginning CV of Bond	Interest Expense (CV x Initial yield) [column (a) x 10%]	Discount Amortized (Interest Expense less Coupon Interest) [column (b) – \$60]	Bond Discount Balance at Year- End	Ending CV of Bond (Beginning CV + Discount Amortized) [column (a) + column (c)]
2025	\$ 930.5	\$ 93.1	33.1	$69.5 - 33.1 = 36.4$	\$ 963.6
2026	963.6	96.4	36.4	$36.4 - 36.4 = 0$	1,000

Example 3 Illustrated via T-Accounts

Bond Payable

	(1) \$1,000 (31/12/24)
	\$1,000 (31/12/25)
(4) \$1,000 (31/12/26)	
	\$0 (31/12/26)

Bond Discount

(1) \$69.5 (31/12/24)	
	(2) \$33.1 (31/12/25)
\$36.4 (31/12/25)	
	(3) \$36.4 (31/12/26)
\$0 (31/12/26)	

Interest Expense (Retained Earnings)

(2) \$93.1 (31/12/25)	
(3) \$96.4 (31/12/26)	

Cash

(1) \$930.5 (31/12/24)	
	(2) \$60 (31/12/25)
	(3) \$60 (31/12/26)
	(4) \$1,000 (31/12/26)
	\$189.5

Note that cumulative interest expense of \$189.5
= cumulative net cash outflows of \$189.5

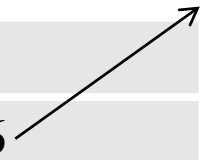
Example 4

\$1,000 6% bond due 31/12/2026 priced to yield 4% and interest compounded and paid annually.

- Annual coupon interest = $\$1,000 \times 6\% = \60 in each of 2025 and 2026.
- Issue price of bond = $\$60 \times PVANN(2, 4\%) + \$1,000 \times PV(2, 4\%)$
 $= \$60 \times 1.89 + 1,000 \times 0.925$
 $= 113.1 + 924.5 = \$1,037.6$
- Bond is issued at a *premium* of \$37.6, Why?
- When *effective interest rate* < *coupon rate*, the bond's issue price is *greater* than its face value!
- Carrying value of bond at 31/12/24 = face value of \$1,000 + unamortized premium of \$37.6 = \$1,037.6

Premium on bond has a normal credit balance so that its amortization is a debit!

(1) Cash	\$1,037.6	
(1) Premium on Bond		37.6
(1) Bond Payable		\$1,000.0



Example 4 cont'd

- *Cumulative interest expense* recognized by the firm over 2 years = total coupon interest payments of \$120 less premium of \$37.6 = \$82.4
- Interest expense = coupon interest – premium amortized, so that when a bond is issued at a premium, ***interest expense recognized is lower than the coupon interest payment by the amount of premium amortized!***

(2) Interest Expense [2025]	\$41.6 (= 4%*1,037.6)	
(2) Premium amortized (plug)	18.4	
(2) Cash		\$60 (=6%*\$1000)

- Unamortized premium at 31/12/2025 = \$37.6 – 18.4 = \$19.2, so that carrying value of bond at 31/12/2025 = \$1,037.6 – 18.4 = \$1,019.2

(3) Interest Expense [2026]	\$40.8 (= 4%*1,019.1)	
(3) Premium amortized (plug)	19.2	
(3) Cash		\$60 (=6%*1,000)

Example 4 cont'd

- Note that the unamortized premium at 31/12/2026, just after the second and final interest payment is $\$37.6 - 18.4 - 19.2 = \0 .
- Carrying value of bond at 31/12/2026 = $\$1,019.2 - 19.2 = \$1,000$ = face value, which is paid off at maturity 31/12/2026 as follows:

(4) Bond Payable	\$1,000	
(4) Cash		\$1,000

- Interest expense can also be calculated using the following *bond amortization schedule*:

	(a)	(b)	(c)	(d)	(e)
Year	Beginning CV of Bond	Interest Expense (CV x Initial Yield) [column (a) x 4%]	Premium Amortized (Interest Expense less Coupon Interest) [\$60– column (b)]	Premium Balance at Year-End	Ending CV of Bond (Beginning CV –Premium Amortized) [column (a) – column (c)]
2025	\$1,037.6	\$ 41.6	18.4	$37.6 - 18.4 = 19.2$	\$ 1,019.2
2026	\$1,019.2	40.8	19.2	$19.2 - 19.2 = 0$	1,000

Example 4 Illustrated via T-Accounts

Bond Payable

	(1) \$1,000 (31/12/24)
	\$1,000 (31/12/25)
(4) \$1,000 (31/12/25)	
	\$0 (31/12/26)

Bond Premium

	(1) \$37.6 (31/12/24)
(2) \$18.2 (31/12/25)	
	\$19.2 (31/12/25)
(3) \$19.2 (31/12/26)	
\$0 (31/12/26)	

Interest Expense (Retained Earnings)

(2) \$41.8 (31/12/25)	
(3) \$40.6 (31/12/26)	

Cash

(1) \$1,037.6 (31/12/24)	
	(2) \$60 (31/12/25)
	(3) \$60 (31/12/26)
	(4) \$1,000 (31/12/26)
	\$82.4

Note that cumulative interest expense of \$82.4
= cumulative net cash outflows of \$82.4.

Retirement of Long-Term Debt

- Recall that, if we retire a bond with face value F on schedule (i.e., at maturity), the journal entry is simply:

Bond Payable	\$ F	
Cash		\$ F

- But what if we retire a bond early, i.e., before maturity?
 - Difference between the *reacquisition price* and *carrying value* of the bond results in gain or loss on retirement.
 - Reacquisition price is market price of bond (including any call premium) plus any pre-payment penalties and fees.
 - If the reacquisition price of the bond exceeds its carrying value, then there is a loss on the retirement of the *bond* . Otherwise, there is a gain on retirement of the bond.

Retirement of Long-Term Debt, Example

- Refer to Example 3 of this handout. Suppose that on 31/12/2025, i.e., a year after issuance and *right before* the payment of the first coupon interest, the borrower repurchases the bonds for cash from the market at a market price of \$982. The fees associated with the repurchase of \$15 were paid in cash. Calculate the gain/loss on debt retirement. Show the journal entry to recognize the bond retirement.